MTH318

Examination - July Semester 2025

Advanced Calculus

Friday, 21 November 2025 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FIVE (5)** printed pages (including cover page).

2. You must answer **ALL** questions

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

Determine whether the given series is absolutely convergent, conditionally convergent or divergent. Justify your answer.

Question 1a

$$\sum_{n=1}^{\infty} \frac{3^n}{2^n + n}$$

(5 marks)

Question 1b

$$\sum_{n=1}^{\infty} (-1)^n \frac{2n}{n^3 + 1}$$

(5 marks)

Question 1c

$$\sum_{n=1}^{\infty} \frac{1}{2}(2n + (-1)^n)$$

(5 marks)

Question 2

Discuss the convergence of the given series or sequence of functions by stating whether it converges uniformly, converges pointwise or diverges on its domain. Justify your answer.

Question 2a

$$\sum_{n=1}^{\infty} \frac{(1+x)^n}{(1-x)(1-x^2)\cdots(1-x^n)}, \text{ where } 0 < x < 1.$$

(10 marks)

Question 2b

$$\left\{ \frac{2nx + (-1)^n \sin(nx)}{n} \right\}_{n=1}^{\infty}, \text{ where } x \in \mathbb{R}.$$

(10 marks)

Question 3

Determine whether each of the following statements are true or false. Give a proof or a counterexample, as appropriate.

Question 3a

If a series of numbers

$$\sum_{n=1}^{\infty} a_n$$

converges and a sequence of numbers $\{b_n\}_{n=1}^{\infty}$ converges to $\lim_{n\to\infty} b_n = 0$, then

$$\sum_{n=1}^{\infty}(a_n + b_n)$$

converges.

(7 marks)

Question 3b

For each $n = 1, 2, 3, \ldots$, let $f_n(x)$ be an increasing function in $x \in [0, 1]$. If

$$\sum_{n=1}^{\infty} f_n(0) \text{and} \sum_{n=1}^{\infty} f_n(1)$$

are both absolutely convergent series of numbers, then

$$\sum_{n=1}^{\infty} f_n(x)$$

converges absolutely on $[0, 1]$.

(6 marks)

Question 3c

Let $\{f_n(x)\}_{n=1}^{\infty}$ and $\{g_n(x)\}_{n=1}^{\infty}$ be sequences of functions on $[0, 1]$ such that $|f_n(x)| < g_n(x)$ for all $x \in [0, 1]$, and for all $n = 1, 2, 3, \ldots$.

If $\{g_n(x)\}_{n=1}^{\infty}$ converges pointwise on $[0, 1]$, then $\{f_n(x)\}_{n=1}^{\infty}$ converges uniformly on $[0, 1]$.

(7 marks)

Question 4

For $n = 1, 2, 3, \ldots$, let $f_n(x) = (-1)^n \dfrac{2}{n + x^2}$, for all $x \in \mathbb{R}$.

Question 4a

Show that

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on $\mathbb{R}$. Justify your answer fully.

(7 marks)

Question 4b

Calculate

$$\int_{-\infty}^{\infty} S(x)dx,$$

where

$$S(x) = \sum_{n=1}^{\infty} f_n(x)$$

is the uniform limit. You may leave your answer as a series of numbers.

(7 marks)

Question 4c

Show that, for each fixed $x_0 \in \mathbb{R}$, the series of numbers,

$$\sum_{n=1}^{\infty} f_n(x_0) = \sum_{n=1}^{\infty} (-1)^n \frac{2}{n + {x_0}^2}$$

does not converge absolutely.

(6 marks)

Question 5

Compute the radius of convergence and the interval of convergence for each of the following power series. Show all your workings completely.

Question 5a

$$\sum_{n=1}^{\infty}\left(1+\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{2n+1}\right)(2x-1)^n\,,$$

(9 marks)

Question 5b

$$\sum_{n=1}^{\infty}(\sqrt{n+1}-\sqrt{n})^n x^n\,,$$

(8 marks)

Question 5c

$$\sum_{n=1}^{\infty}\frac{n^{n+1}}{3^n}(3x+2)^n\,.$$

(8 marks)

----- END OF PAPER -----